1. Set up the system by fixing a specific position and velocity for each part.
2. Evolve the system using Newton's laws of motion.

That's it. The devil is in the details, of course. Some classical systems can have a lot of moving pieces.

In contrast, the rules of standard textbook quantum mechanics come in two parts. In the first part, we have a structure that exactly parallels that of the classical case. Quantum systems are described by wave functions rather than by positions and velocities. Just as Newton's laws of motion govern the evolution of the state of a system in classical mechanics, there is an equation that governs how wave functions evolve, called *Schrödinger's equation*. We can express Schrödinger's equation in words as: "The rate of change of a wave function is proportional to the energy of the quantum system." Slightly more specifically, a wave function can represent a number of different possible energies, and the Schrödinger equation says that high-energy parts of the wave function evolve rapidly, while low-energy parts evolve very slowly. Which makes sense, when we think about it.

What matters for our purposes is simply that there is such an equation, one that predicts how wave functions evolve smoothly through time. That evolution is as predictable and inevitable as the way objects move according to Newton's laws in classical mechanics. Nothing weird is happening yet.

The beginning of the quantum recipe reads something like this:

Rules of Quantum Mechanics (Part One)

1. Set up the system by fixing a specific wave function Ψ.
2. Evolve the system using Schrödinger's equation.

So far, so good—these parts of quantum mechanics exactly parallel their classical predecessors. But whereas the rules of classical mechanics stop there, the rules of quantum mechanics keep going.

All the extra rules deal with measurement. When you perform a measurement, such as the position or spin of a particle, quantum mechanics says there are only certain possible results you will ever get. You can't